

Sample Admissions Test

Paper 1

Sample Set 3 (2026)

Time allowed: 1 hour 15 minutes

There are 8 questions. Answer all of them.

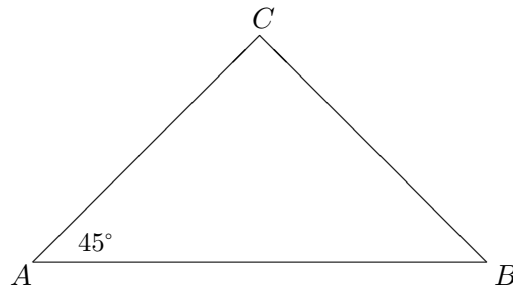
Calculators are not permitted.

- 1 Find the complete set of values of x for which

$$\frac{x^2 - 4}{x + 1} \leq 0$$

- A** $x \leq -2$ **B** $-2 \leq x < -1$ or $x \geq 2$ **C** $x \leq -2$ or $-1 < x \leq 2$
D $-1 < x \leq 2$ **E** $x \geq 2$

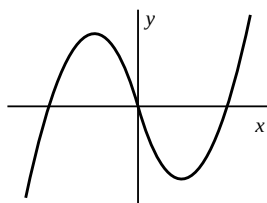
- 2 The diagram shows triangle ABC with $AB = 9/\sqrt{2}$ and angle $BAC = 45^\circ$.



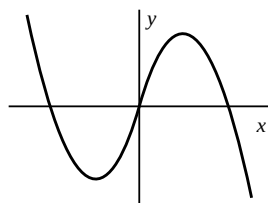
What is the area of triangle ABC ?

- A** $\frac{81}{4}$ **B** $\frac{81}{2}$ **C** $\frac{81\sqrt{2}}{4}$
D 81

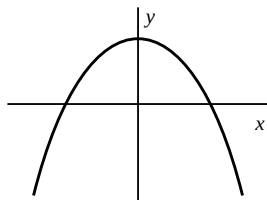
- 3** Which one of the following could be the graph of $y = x^3 - x$?



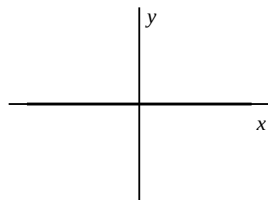
A



B



C



D

BLANK PAGE

4 Evaluate

$$\sum_{k=1}^{10} \log_2 \left(\frac{k+1}{k} \right)$$

- A** $\log_2 11$ **B** $\log_2 10$ **C** $1 + \log_2 11$
D 11 **E** $\log_{10} 2$

5 Consider the following statements about the function $f(x) = |x - 1| - 2$:

- I** $f(-1) = 0$
II $f(x) \geq -2$ for all real x
III the equation $f(x) = 1$ has exactly two solutions

Which of these statements is/are true?

- A** none of them **B** I only **C** II only
D I and II only **E** I, II and III

6 Given that $\int_0^a (3x^2 - 2x)dx = 4$ where $a > 0$, find the value of a .

A 1 **B** 2 **C** $\frac{1+\sqrt{17}}{2}$

D $\sqrt{3}$ **E** 4

7 The sequence u_n is defined by $u_1 = 2$ and $u_{n+1} = \frac{u_n}{1+u_n}$ for $n \geq 1$.
What is the value of u_{10} ?

- A** $\frac{2}{19}$ **B** $\frac{1}{10}$ **C** $\frac{2}{21}$
D 2^{-10} **E** $\frac{1}{29}$

- 8 The table shows some values of the function f .

x	0	1	2	3
$f(x)$	1	3	5	7

Which one of the following could be $f(x)$?

- A** $2x + 1$ **B** $x^2 + 1$ **C** 3^x
D $\sqrt{x + 1}$ **E** $\frac{6}{x+1}$

END OF TEST